Allamaprabhu S Ani

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SUMMARY

PhD researcher in computational mechanics and machine learning, building GPU-native, differentiable physics simulators in PyTorch. Lead author of the 2026 *Engineering Fracture Mechanics* review on ML for fracture (with EPFL and University of Florida). Production-software discipline: torch_pf_solver, a PyTorch package with end-to-end autograd through a staggered FEM solver, reproducible Slurm pipelines, cross-validated against five reference codes. Targeting ML research engineer / ML-for-science roles.

SKILLS

Programming: Python, PyTorch (autograd, CUDA), JAX, NumPy, SciPy, scikit-learn, pandas, MATLAB, C++ (read), Bash, Git.
Machine Learning: Neural operators (DeepONet, FNO), PINNs, differentiable simulation, inverse problems, L-BFGS, adjoint methods.
Scientific Computing: GPU acceleration, sparse linear algebra, finite element method, phase-field fracture, mixed-precision numerics.
Infrastructure: Slurm HPC, Linux, reproducible packaging, latexmk, Sphinx, GitHub Actions.
Engineering Tools: Akantu (C++), FEniCSx, COMSOL, MSC Nastran/Patran, Abaqus, Altair Hypermesh, ANSYS.

EXPERIENCE

Doctoral Researcher, Computational Mechanics and ML — Jul 2023 – Jun 2026 (expected)
City, St George's, University of London & Queen Mary University of London (CEMS Lab)

- Built torch_pf_solver: PyTorch-native phase-field fracture solver with end-to-end autograd through a staggered solver, scaling to 1M+ DOF on a single GPU.
- Implemented gradient-based inverse problems for material parameter recovery, spatial field reconstruction, and defect localisation; benchmarked against finite-difference and MCMC baselines.
- Achieved 10–20$\times$ sparse-solve speedup over a C++ reference (Akantu) on quasi-static benchmarks; cross-validated dynamics on Kalthoff, SENT, Borden.
- Reproducible Slurm pipelines on Hyperion2; local dev on Mac CPU and WSL CUDA with deterministic float64.

Visiting Researcher, Computational Solid Mechanics — 2025
LSMS Lab, EPFL (host: Prof. J.-F. Molinari)

- Co-authored Engineering Fracture Mechanics 2026 review of ML for computational fracture; open access, growing citation count.

Founder and Tool Developer — 2020 – 2021
Aeroknacks India Ltd

- Shipped a library of validated structural-analysis tools (BJSFM, Cozzone, lug strength, fastener load transfer, buckling, modal, ABD) to an aerospace engineering firm; built on Bruhn/Niu/ESDU references and cross-validated in MSC Nastran/Patran.

EDUCATION

PhD, Mechanical Engineering, City, St George's, University of London (Jul 2023 – Jun 2026 exp.). Thesis: GPU-native differentiable phase-field fracture with neural-operator constitutive laws. Supervisor: Dr S.A. Ponnusami.
MTech, Design and Manufacturing, NIT Silchar (2021–2023). CGPA 9.88/10. Best Student Award.
BE, Mechanical Engineering, GM Institute of Technology, VTU (2016–2020). CGPA 8.47/10.

PUBLICATIONS

- **Ani, A.S.**, Nakka, R., Subhash, G., Molinari, J.-F., Ponnusami, S.A. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics* 332, 111778. Open access.
- **Ani, A.S.**, Deoghare, A.B. (2024). Leveraging Machine Learning for Enhanced Fatigue Life Prediction in Aluminium Alloys. *Lecture Notes in Mechanical Engineering*, Springer.

PROJECTS AND OPEN SOURCE

bjsfm | github.com/allamaprabhuani/bjsfm – Anisotropic stress field around a hole in a composite plate (Lekhnitskii, de Jong) in Python.
geometric-transformations | github.com/allamaprabhuani/geometric-transformations – 2D/3D affine transforms with MATLAB Live Scripts.
allamaprabhuani.github.io – research notes, ML tutorials, blog.

AWARDS

Yeoman & Travelling Scholarship, Worshipful Company of Tin Plate Workers (2026, 1 of 3 citywide). Dean's Award for Outstanding Teaching Support, City St George's (2026). Associate Fellow of the Higher Education Academy (AFHEA, 2024). Funded PhD Studentship, City St George's (2023). Best Student Award, MTech NIT Silchar (2023).